# Extended reading — Week 19: DPO and the Preference Zoo

October 6, 2026

## How to read this list

Each paper below: full citation, one-paragraph summary, a *Why it matters for this week* note, and one annotated key passage with brief commentary.

## 1. Rafailov et al. 2023 — Direct Preference Optimization

**Citation.** R. Rafailov, A. Sharma, E. Mitchell, S. Ermon, C. D. Manning, C. Finn, *Direct Preference Optimization: Your Language Model is Secretly a Reward Model*, arXiv:2305.18290, NeurIPS 2023.

**Summary.** Derives, in three pages, the equivalence between the KL-regularised RL objective and a closed-form supervised contrastive loss. The reward model can be re-parametrised through the policy as $r_\theta = \beta \log(\pi_\theta / \pi_{\text{ref}})$; the partition function cancels in pairwise comparisons; what remains is a logistic-regression-style loss on the policy. Empirically, DPO matches or beats PPO-RLHF on summarisation and dialogue at a fraction of the compute and engineering cost.

**Why it matters for this week.** This is the paper the whole week is built around. The 4-step derivation in our notes is their Theorem 1.

**Annotated key passage.**

> *"This change of variables enables us to express the RL objective in terms of the policy directly, avoiding the need to learn an explicit reward model... The resulting optimization problem can be solved using a standard supervised learning algorithm, without the need for RL."*

The two clauses are operationally massive. "Avoiding the need to learn an explicit reward model" kills three of the four models in the RLHF pipeline. "Without the need for RL" kills the remaining engineering surface (rollouts, GAE, clipping, KL controller, value head). The community's adoption of DPO within twelve months was almost entirely a story about this engineering simplification.

## 2. Azar et al. 2024 — IPO and the ΨPO framework

**Citation.** M. G. Azar, M. Rowland, B. Piot, D. Guo, D. Calandriello, M. Valko, R. Munos, *A General Theoretical Paradigm to Understand Learning from Human Preferences*, arXiv:2310.12036, AISTATS 2024.

**Summary.** Generalises DPO into ΨPO, a family parametrised by the choice of link function $\Psi$. Shows DPO's logistic link saturates as preferences become deterministic, collapsing the implicit KL constraint. Proposes IPO — a squared loss, $\Psi(u) = (u - 1/(2\beta))^2$ — whose unique optimum has a *finite* margin and avoids the saturation pathology.

**Why it matters for this week.** The link-function plot in the notes (`fig_links.pdf`) is theirs. IPO is the cleanest theoretical advance over DPO; KTO, ORPO, and SimPO are best understood as alternative *structural* (not link-function) choices on top of this framework.

**Annotated key passage.**

> *"As the preferences become more deterministic, the empirical optima of DPO drive the implicit reward gap to infinity, regardless of the value of $\beta$. IPO instead has a finite optimum..."*

The two-sentence rebuke of DPO. The IPO "finite optimum" is also the reason IPO does *not* need a KL leash: the squared loss provides its own regularisation by pinning the margin to a finite target.

## 3. Ethayarajh et al. 2024 — KTO

**Citation.** K. Ethayarajh, W. Xu, N. Muennighoff, D. Jurafsky, D. Kiela, *KTO: Model Alignment as Prospect Theoretic Optimization*, arXiv:2402.01306, 2024.

**Summary.** Frames alignment through Kahneman–Tversky prospect theory: human preferences are noisy and asymmetric — gains relative to a reference are felt less strongly than equal-sized losses. KTO replaces the BT pairwise model with a per-sample *utility* that is concave on gains and convex on losses, allowing training on *unpaired* thumbs-up / thumbs-down labels.

**Why it matters for this week.** Production data is rarely in clean pairwise form; KTO is the practical way to use the messy, unpaired feedback that the deployed product accumulates. The synthetic insurance matching story in the notes lives here.

**Annotated key passage.**

> *"KTO does not need preferences — only a binary signal of whether an output is desirable or undesirable for a given input. This makes it far easier to use in the real world, where this kind of data is much more abundant."*

The pragmatic case for KTO in one sentence. The deeper theoretical contribution is that the prospect-theory utility is *not* an arbitrary heuristic: it falls out of empirically calibrated human-decision data, giving KTO a behavioural-science foundation that DPO lacks.

## 4. Hong et al. 2024 — ORPO

**Citation.** J. Hong, N. Lee, J. Thorne, *ORPO: Monolithic Preference Optimization without Reference Model*, arXiv:2403.07691, EMNLP 2024.

**Summary.** A single-stage method that fuses SFT NLL with a log-odds-ratio penalty on preference pairs, eliminating the reference policy entirely. The result: a one-stage pipeline that trains on the same data DPO uses, at half the memory, with a small quality bump on AlpacaEval and IFEval.

**Why it matters for this week.** ORPO is the natural contrast to DPO on the "do I need a reference?" axis. It also forces us to grapple with numerically stable log-odds-ratio computation — a subtle bit of practical engineering exposed in `losses._log1mexp`.

**Annotated key passage.**

> *"Despite ORPO being a single-step training method, models trained with ORPO surpass much larger Llama-2-Chat and Zephyr variants trained with the conventional multi-stage RLHF pipeline."*

The bold empirical claim. With caveats: ORPO's single-stage nature couples preference data quality to SFT data quality very tightly. On clean curated data it shines; on messy production data it can amplify bias.

## 5. Meng, Xia, Chen 2024 — SimPO

**Citation.** Y. Meng, M. Xia, D. Chen, *SimPO: Simple Preference Optimization with a Reference-Free Reward*, arXiv:2405.14734, NeurIPS 2024.

**Summary.** The most ergonomic DPO variant: reference-free, length-normalised, and with a target margin parameter $\gamma$. The implicit reward is the average per-token log-probability. Gemma-2-9B + SimPO hits 72.4% length-controlled win rate on AlpacaEval-2 without RLHF.

**Why it matters for this week.** SimPO is the recommended starting baseline for DPO-family work in 2026: it requires no reference model, has built-in length-normalisation (which by itself solves one of DPO's biggest headaches), and has only two hyper-parameters.

**Annotated key passage.**

> *"We propose SimPO, a simple yet effective approach to preference optimization that uses the average log probability of a sequence as the implicit reward. This eliminates the need for a reference model and naturally aligns with the metric used at inference time."*

The "aligns with inference" point is subtle and important. DPO's implicit reward is the *sum* of token log-probabilities, but decoding ranks candidates by the length-normalised score (perplexity). SimPO simply aligns training with the inference objective.

## 6. Liu et al. 2024 — Iterative Length-Regularized DPO

**Citation.** W. Liu, K. Bai, Y. Ren, Y. Bian, J. Bian, Z. Wang, *Iterative Length-Regularized Direct Preference Optimization: A Case Study on Improving 7B Language Models to GPT-4 Level*, arXiv:2406.11817, 2024.

**Summary.** Two ideas: (i) iterate DPO by regenerating preferences with the latest policy, refreshing the implicit-reward target; (ii) add a length-regularization term to the DPO loss to combat length inflation. The combination pushes Llama-3-8B to GPT-4-level on AlpacaEval, a striking result given the model size.

**Why it matters for this week.** Bridges DPO and the iterative-DPO / self-rewarding literature of week 20. Also the most direct piece of evidence that length bias — the failure mode SimPO addresses architecturally — can be fixed equally well at the loss level.

**Annotated key passage.**

> *"Iterative DPO without length regularization can win the AlpacaEval-2 game on raw win rate while losing the length-controlled win rate by a wide margin. This is a clear signal that the policy is being rewarded for verbosity rather than quality."*

The honest empirical observation that AlpacaEval-2's length-controlled metric was *designed* to expose: optimisers hunt the easiest path to winning, and verbosity is the easiest path under sum-log-prob accounting.

## 7. Xu et al. 2024 — CPO for machine translation

**Citation.** H. Xu, A. Sharaf, Y. Chen, W. Tan, L. Shen, B. Van Durme, K. Murray, Y. J. Kim, *Contrastive Preference Optimization: Pushing the Boundaries of LLM Performance in Machine Translation*, arXiv:2401.08417, 2024.

**Summary.** Introduces CPO — SFT NLL on the chosen translation plus a contrastive log-ratio between chosen and rejected. No reference model, halving the memory footprint vs. DPO. Achieves state-of-the-art on multiple WMT language pairs at the time of publication.

**Why it matters for this week.** The clearest empirical case for the *halfway-house* family (SFT + contrastive, no reference): MT is a domain where the SFT signal dominates and the preference signal is polish.

**Annotated key passage.**

> *"By removing the reference model, CPO not only reduces memory requirements but also liberates the optimisation from the constraint of having to stay close to a possibly suboptimal reference."*

The two-edged blade of reference-free training: cheaper, but you lose the regularising effect of the SFT anchor. CPO's solution — include the SFT NLL term — is one of three reasonable substitutes (the others being a target margin like SimPO's $\gamma$ and a length normaliser).

## 8. Yuan et al. 2024 — Self-Rewarding Language Models

**Citation.** W. Yuan, R. Y. Pang, K. Cho, X. Li, S. Sukhbaatar, J. Xu, J. Weston, *Self-Rewarding Language Models*, arXiv:2401.10020, 2024.

**Summary.** The same LLM plays generator, judge, and learner. LLM-as-a-judge prompting produces preference scores; DPO updates the policy on those scores; the loop iterates. Three iterations of self-rewarding Llama-2-70B beats Claude 2, Gemini Pro and GPT-4-0613 on AlpacaEval-2.

**Why it matters for this week.** Bridges DPO and the iterative-DPO family of week 20. The DPO loss is unchanged; what changes is the preference data source. The result is striking but the failure mode — self-confirmation drift — is also striking.

**Annotated key passage.**

> *"To use the model itself as a reward judge, we use prompts that instruct it to assess responses, providing a numerical score that we can use as a reward signal for further training. Crucially, the model's ability to judge improves alongside its ability to generate."*

The optimistic claim. The pessimistic one is that the model's biases as a judge co-evolve with its biases as a generator, producing a runaway feedback loop. Both claims are true to some extent; the equilibrium depends on data freshness and the quality of the initial judge.

## Pointers for further depth

- Rainbow PO — arXiv:2410.04203, 2024. A unifying survey that places DPO, IPO, KTO, ORPO, SimPO, CPO inside a single "link function × normalisation" lattice.
- DPO survey — arXiv:2410.15595, 2024–2025. A comprehensive survey of the DPO family up to late 2024.
- Tang et al. 2024 — *Generalised Preference Optimization*, arXiv:2402.05749. An importance-weighted analysis of when DPO and explicit-RM-RL gradients agree.
- Park et al. 2024 — *Disentangling Length from Quality in DPO*, arXiv:2403.19159. A clean empirical decomposition of why DPO inflates length.
- Iterative DPO Study — arXiv:2503.12854, 2025. Comprehensive study of iterative-DPO design choices: how often to refresh, how many pairs per iteration, when to stop.
- Tülu-3 — arXiv:2411.15124, 2024. The clearest recipe for combining SFT, DPO, and RLVR end-to-end.

## Sourced from

`research/llm_rl_sota.md` §2 (the DPO zoo) and §6 (iterative / online DPO).